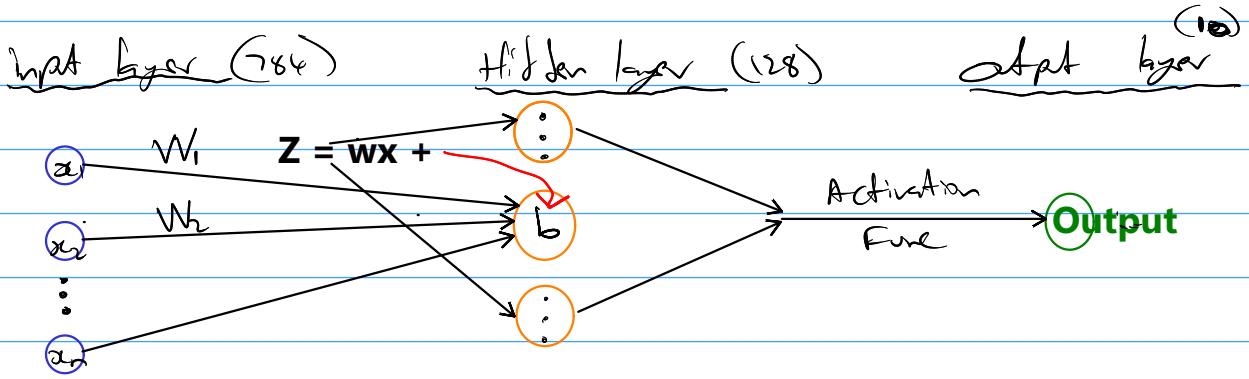
Understanding forward pass



Forward propagation is a process where input data, in this case it an array of values. which are pixels ranging from 0 (black) & 255 (white) and some grays in between.

e.g. image 1 : [0, 0, 0, 255, ..., 128, 0]

passes through each layer of a neural networks to produce an output, which is just a probability value to classify each input data as a number from 0 - 9.

it does this by transforming raw inputs into prediction using weights, biases which are agured during the model training phase and an activation functions. of which for this forward pass its softmax & ReLu reason for softmax is multiple classification / labeling (0 - 9). And ReLu adds non-linearity to help network learn flatres.

NB: The forward pass is inference, not training. it uses fixed, pretrained weight and biases to prdue probabilities & nothing gets updated.

For more information read below.

https://www.geeksforgeeks.org/deep-learning/forward-propagation-in-neural-networks/

https://www.geeksforgeeks.org/deep-learning/relu-activation-function-in-deep-learning/

https://www.geeksforgeeks.org/deep-learning/the-role-of-softmax-in-neural-networks-detailed-explanation-and-applications/